Detecting Unanswerable Visual Questions Using Image-Text Fusion

Multi Modal Analysis- Assignment 2.2

Barathkumar Palanisamy (1952609)

GitHub repository: https://github.com/BarathPalanisamyy/vizwiz-answerability-fusion

1. Introduction

Visual Question Answering (VQA) is a computer vision application that also integrates Natural Language Processing (NLP) to answer questions about images. But not all the questions can be answered based on the visual information. Images may be out of focus, poorly framed, blocked or not answer the question. In accessibility applications, individuals may rely on automated systems to comprehend their environment if they have visual impairments, which makes this problem especially crucial.

In this project, it is examined if the image and text features help to improve the ability to detect the unanswerable questions, present in the VizWiz-VQA dataset (Gurari et al., 2018). The task is considered as a binary classification problem with 0 denoting "answerable" and 1 denoting "unanswerable".

The primary goal is to see how unimodal baselines perform when fusing with various methods and investigate their performance, generalisation, efficiency and potential biases. The experiments are conducted with early fusion, late fusion, cross-attention, gated fusion and early fusion with CLIP. Special interest is shown in whether more complex fusion techniques can offer significant gains over simpler techniques.

2. Dataset and Preprocessing

2.1 Dataset Description

VizWiz-VQA is a collection of images and questions gathered from real-life visual disability users. Many images are difficult to handle, e.g. due to lighting conditions or due to blurring or due to only part of the object being shown, as is common in VQA data sets.

The dataset comprises 32,842 image-question combinations, divided into training, validation and test sets, as shown in Table 1.

Dataset split	Samples	Unanswerable proportion
Training	20,523	27.0%
Validation	4,319	32.1%
Test	8,000	26.8%
Total	32,842	

Table 1. Distribution of answerable and unanswerable questions across the VizWiz dataset splits.

The labels of the original answerability annotations were converted into binary labels. Test annotations were checked against the image identifiers and question wording for possible accuracy before evaluation. The matching of all images with their corresponding files was achieved successfully.

The class distribution shows that there are more answerable questions than unanswerable questions in the dataset. So, to represent performance on the unanswerable class, accuracy alone would not be sufficient

2.2 Preprocessing

Images were converted to RGB, resized to 224 × 224 pixels and normalised using the corresponding pretrained encoder settings. The questions were tokenised with BERT with the max length of 40 tokens. The image and text pre-processing algorithms were implemented in CLIP.

To analyze further, grayscale images were used to estimate image sharpness and the training split was used to calculate question frequency. Questions that appeared once in the training set were considered rare, including the questions that were not seen during training. This eliminated the need to set up validation or test question frequencies for group definitions.

The same partitions were used throughout the experiments to facilitate comparison of the models.

3. Methodology

3.1 Feature Extraction

The implementation relies on pre-trained encoders for image and question encoding in numerical features.

The ViT-B/16 model is Vision Transformer which outputs a 768-dimensional image representation of the entire image and a set of features at each patch level. The patch representations are pooled into 49 spatial regions for cross attention. ViT performs transformer like processing on image patches (Dosovitskiy et al., 2021).

BERT-base produces question embeddings and token representations for context. The bidirectional attention mechanism helps it to grasp the relationship between words in a question (Devlin et al., 2019).

Another set of image and text embeddings are created from the CLIP approach, which is learned from contrastive image-text training (Radford et al., 2021).

All pretrained encoders were frozen and classification and fusion layers only were trained. This simplified the calculation and enabled a comparison to be made of the different fusion strategies

3.2 Baselines and Fusion Architectures

Two unimodal baselines were created. The question-only model uses BERT features, while the image-only model uses ViT features. These set the pieces of information for every modality alone.

Five multi modal configurations were tested:

- **Early fusion:** Concatenates image and question embeddings before classification.
- **Late fusion:** Combines unimodal prediction probabilities using validation-selected weights.
- **Cross-attention:** Uses question tokens to attend to image-region features, following transformer attention principles (Vaswani et al., 2017).
- **Gated fusion:** Learns the contribution of each modality and uses modality dropout during training.
- **CLIP early fusion:** Combines pretrained CLIP image and text embeddings through a classification head.

The advantage of early fusion is that it offers a simple fusion at the feature level, while the cross-attention models offer more detailed interactions between the question tokens and image regions. Gated fusion enables the network to adapt its contributions from each modality. These methods can be compared to find out if architectural complexity increases answerability detection or not.

4. Experimental Setup and Evaluation

The models were built with PyTorch and trained with binary cross-entropy loss and AdamW optimiser (Loshchilov & Hutter, 2019). The learning rate was 0.001, batch size was 256 and weight decay was 0.0001. Early stopping was implemented after 3 epochs with no sign of improvement on the validation loss and training was stopped after 15 epochs.

Three random seeds (0, 1 and 2) were used for each configuration. Early stopping, late-fusion weighting and classification-threshold selection were done by using validation data. The test set was held back for the final evaluation.

The major measure used was macro-F1 that weighted both classes equally. AUROC is used to evaluate discrimination over thresholds and AUPRC is used to evaluate precision recall over the unanswerable class. The accuracy, recall and Brier score were also noted.

Paired bootstrap resampling over 500 resamplings provided estimates of 95% confidence intervals for improvements in performance. Further analyses of training validation gaps, of the behaviour of subgroups, of the prediction errors and of the latency of the inferences were performed.

5. Results and Model Comparison

5.1 Overall Classification Performance

Table 2 reports the mean test performance across three random seeds.

Model	Macro-F1	AUROC	AUPRC
Question-only	0.6568	0.7447	0.5256
Image-only	0.6946	0.7737	0.5932
Early fusion	0.7617	0.8663	0.7235
Late fusion	0.7409	0.8459	0.6828
Cross-attention	0.7613	0.8618	0.7132
Gated fusion	0.7540	0.8600	0.7113
CLIP early fusion	**0.7990**	**0.9000**	**0.7783**

Table 2. Average test performance of all seven models across three random seeds.

Both unimodal baselines were outperformed by all multimodal configurations. The CLIP early fusion model performed best on all three metrics, outperforming the image-only baseline by 0.1044 in macro-F1.

The results of the early fusion and cross-attention models were comparable, with macro-F1 scores of 0.7617 and 0.7613, respectively. While cross-attention models better captures the details of the

relationships, it did not beat simple feature concatenation. The other fusion methods also performed better than the late fusion, which did improve over the baselines, though.

The results indicate that a combination of modalities is helpful, and that the quality of fusing pretrained representations may be as crucial as the complexity of the fusion.

5.2 Statistical Comparison

The paired bootstrap analysis supported the improvement over the image-only baseline. In the early fusion of CLIP, the three 95% confidence intervals for macro-F1 improvement were all positive, excluding zero.

The estimated improvement for the seed 0 was around 0.105 with a confidence interval of [0.0933, 0.1174]. This means that the trend is consistently positive when resampling with the bootstrap method

5.3 Generalisation and Overfitting

The CLIP early fusion model provided a training macro-F1 score of 0.8636 and validation macro-F1 score of 0.7914 with a difference of 0.0721. Early fusion had a gap of 0.0666 and cross-attention had a gap of 0.0503.

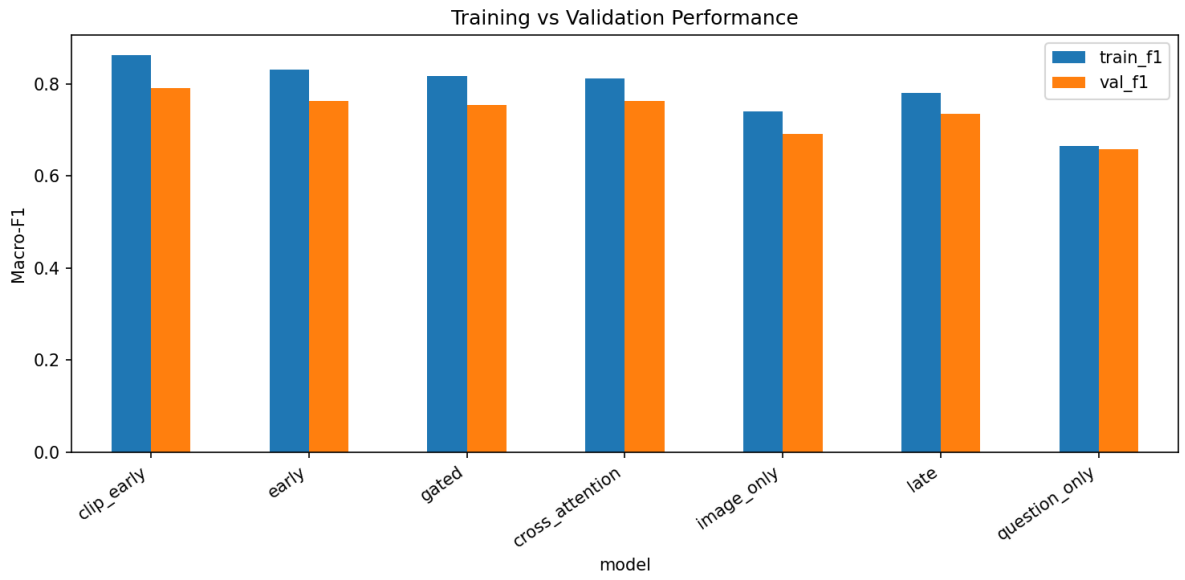


Figure 1. Comparison of training and validation macro-F1 scores across the seven models.

Several fusion models had a better fit to the training data than to the validation data as shown in Figure 1. However, CLIP early fusion scored 0.7990 on test macro-F1, which is near the validation performance. This indicates that there is a reasonable generalisation on the held out test set even though there is a gap between training and validation

6. Error Analysis, Bias and Ethics

6.1 Classification Errors

The confusion matrix in Figure 2 shows the correct and incorrect predictions made by the selected fusion model. It helps identify false positives, where answerable questions are incorrectly rejected, and false negatives, where unanswerable questions are not detected.

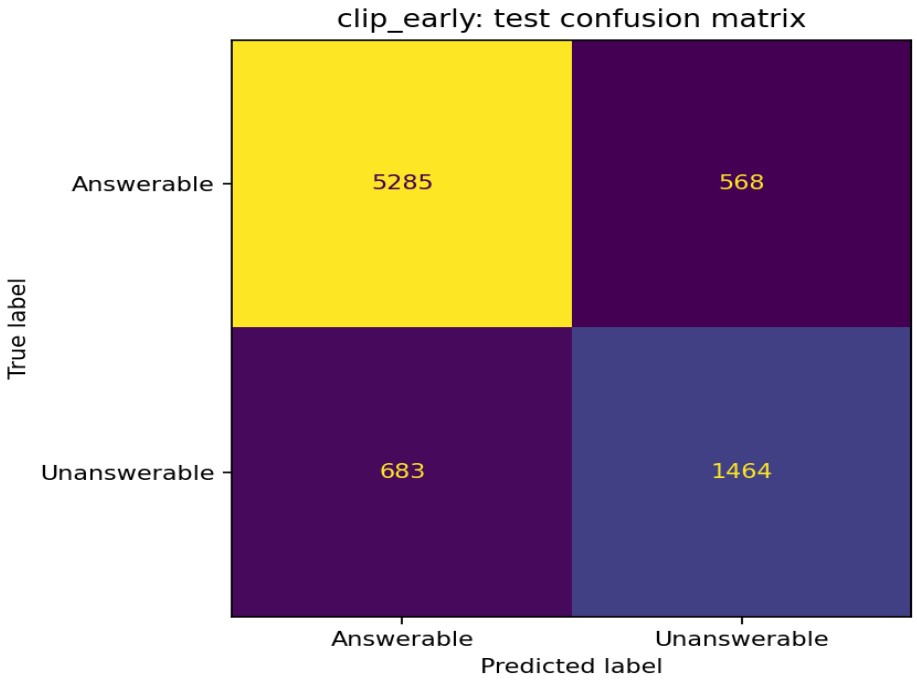


Figure 2. Confusion matrix showing correct and incorrect predictions on the test set for the selected model

The two types of errors are of practical significance. If the VQA system returns a false positive, users will not get information that is useful to them, and if a false negative, a downstream VQA system may try to answer without enough visual information.

6.2 Subgroup Analysis

The performance was analysed on the basis of image sharpness, types of questions and answers. Table 3 shows some selected results for CLIP early fusion.

Subgroup	Samples	Accuracy	Unanswerable recall
Higher sharpness	3,908	0.860	0.571
Lower sharpness	4,092	0.824	0.776
Frequent questions	4,306	0.883	0.702
Rare questions	3,694	0.794	0.713

Table 3. Performance of the CLIP early fusion model across different test data subgroups.

Images of higher sharpness yielded an accuracy of greater than that of images of lower sharpness, whereas that of lower sharpness yielded higher unanswerable recall. There were significant differences in the number of classes between the two, with the lower sharpness images having more unanswerable examples. Hence, the findings are not conclusive evidence for blur improving detection.

Rare questions had a lower accuracy than frequent questions. This could point to the fact they're having trouble transferring to less familiar question patterns, though also class composition.

Different false positive rates were found in different categories for answer type analysis. The macro-F1 for individual answer-type categories was not calculated as there was only one ground-truth class in each of these categories.

6.3 Decision Threshold and Ethical Considerations

The classification threshold impacts the ratio of false positives and negatives. This trade-off is shown in Figure 3 with the predictions made by validation.

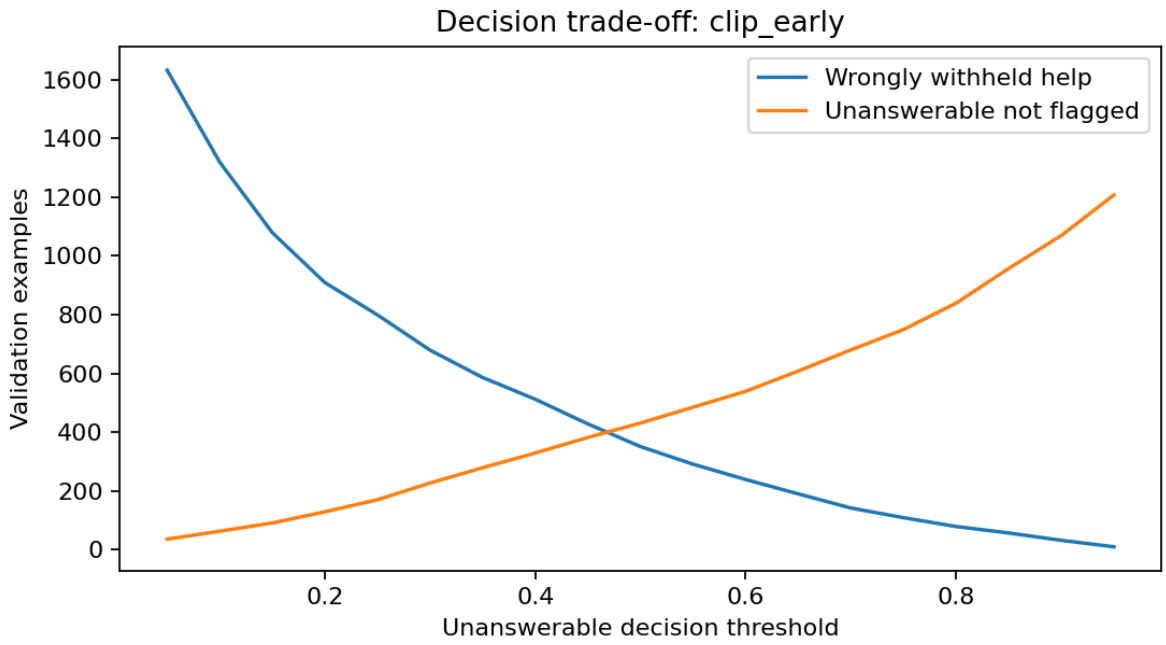


Figure 3. Effect of different classification threshold on false positives and false negatives using validation data.

Raising the threshold for predicting the unanswerable class will decrease the number of rejected answerable cases, but will also increase the number of missed unanswerable cases. The effects of both types of errors should therefore be taken into account when choosing the threshold.

VizWiz images might consist of private areas or individual information, and information management must be taken into consideration. Furthermore, performance may differ across languages, devices, environments and user populations not adequately represented in the dataset.

The system should convey uncertainty for accessibility applications and enable users to retake photos or rephrase questions. Overall, accuracy should not be the only criterion for evaluating the reliability of the system in accessibility applications.

7. Efficiency, Limitations and Reflection

The Notebook computed the inference latency for classifiers-only and the end-to-end. Classifier-only measurements leveraged the cached embeddings while the end-to-end CLIP benchmark also incorporated loading, processing, and classification of local images, as well as classification using CLIP's encoder.

For 50 validation examples, CLIP early fusion achieved a median end-to-end latency of 52.68 ms, mean latency of 57.59 ms and 95th-percentile latency of 130.62 ms. The median was less than the 100ms target, however higher percentile values were over this threshold. No network delays or application overheads were included.

One advantage of the implementation is that seven configurations were tested on three seeds. The experiments were computationally manageable by using frozen encoders, but they were unable to directly adapt their representations to the image conditions of the VizWiz domain.

The findings contradicted the notion that increasingly complex fusions will outperform simpler ones. Cross-attention and gated fusion proved helpful comparisons, but the best performance came from CLIP early fusion. This highlighted the need to test design decisions through experimentation and not assume that a more complex architecture will lead to better results.

Selective encoder fine-tuning, probability calibration and evaluation on other accessibility datasets could be explored in future work.

8. Conclusion

This project investigated multimodal fusion for detecting unanswerable questions in VizWiz-VQA. All fusion approaches outperformed the unimodal baselines, demonstrating the value of combining visual and textual information.

The best overall performance was found to be by CLIP early fusion (macro-F1 = 0.7990, AUROC = 0.9000, and AUPRC = 0.7783). The bootstrap analysis supported the improvement in macro-F1 compared with the image-only baseline, with the 95% confidence intervals excluding zero.

Limitations related to class imbalance and frequency of questions, as well as reliability of deployment were identified in the subgroup and latency analyses. In general, the results indicate that image text fusion is applicable to answerability detection and that careful evaluation needs to be given in accessibility applications.

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